

JavaScript - Unit 1: Introduction to JavaScript

What is JavaScript?

Definition

JavaScript (JS) is a high-level, lightweight, interpreted programming language used to make web pages **interactive and dynamic**.

It works together with **HTML** and **CSS** to create modern web applications.

- **HTML** → Structure of the webpage
- **CSS** → Styling of the webpage
- **JavaScript** → Functionality and Interactivity

Example

```
<button onclick="showMessage()">  
Click Me  
</button>
```

```
<script>  
  
function showMessage(){  
  
    alert("Welcome to JavaScript!");  
  
}  
  
</script>
```

Output

[Click Me]

↓

Popup Message

Welcome to JavaScript!

Features of JavaScript

JavaScript provides many powerful features.

Features

- Lightweight programming language.
 - Easy to learn and use.
 - Object-Oriented Programming support.
 - Platform Independent.
 - Event-driven programming.
 - Supports Client-side and Server-side development.
 - Supports DOM Manipulation.
 - Interpreted language (No compilation required).
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Advantages of JavaScript

- Fast execution in web browsers.
 - Easy integration with HTML and CSS.
 - Supported by all modern browsers.
 - Used in Frontend and Backend development.
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Applications of JavaScript

JavaScript is used in many areas of software development.

Applications

- Website Development
- Web Applications
- Mobile Applications
- Games
- Form Validation
- Interactive Animations
- Online Shopping Websites
- Chat Applications
- Dashboard Applications

Popular Websites Using JavaScript

- Google
- YouTube

JavaScript Syntax

JavaScript statements are written inside the `<script>` tag.

Syntax

```
<script>
```

JavaScript Code

```
</script>
```

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>JavaScript</title>
```

```
</head>
```

```
<body>
```

```
<script>
```

```
document.write("Welcome to JavaScript");
```

```
</script>
```

```
</body>
```

```
</html>
```

Output

Welcome to JavaScript

Internal JavaScript

Definition

When JavaScript code is written **inside the HTML file**, it is called **Internal JavaScript**.

Example

```
<!DOCTYPE html>
<html>

<head>

<title>Internal JavaScript</title>

</head>

<body>

<h2>Internal JavaScript Example</h2>

<script>

document.write("Hello Students!");

</script>

</body>

</html>
```

Output

Hello Students!

External JavaScript

Definition

When JavaScript code is written in a separate **.js** file and linked to an HTML file, it is called **External JavaScript**.

HTML File

```
<!DOCTYPE html>
<html>

<head>
```

```
<script src="script.js"></script>
```

```
</head>
```

```
<body>
```

```
<h2>External JavaScript</h2>
```

```
</body>
```

```
</html>
```

JavaScript File (script.js)

```
document.write("Welcome to External JavaScript");
```

Output

Welcome to External JavaScript

Comments in JavaScript

Definition

Comments are used to explain the code. Comments are ignored during program execution.

Single Line Comment

Syntax

```
// This is a comment
```

Example

```
// Display a message
```

```
document.write("JavaScript");
```

Multi-Line Comment

Syntax

```
/*
```

```
This is  
a multi-line  
comment
```

```
*/
```

Example

```
/*
```

```
Student Details
```

```
*/
```

```
document.write("Hari");
```

Summary of Unit 1

Topic	Description
JavaScript	Programming language used to create interactive webpages
Features	Lightweight, Event-driven, Object-Oriented
Advantages	Fast, Easy, Dynamic, Browser Supported
Applications	Websites, Games, Mobile Apps, Web Apps
Internal JS	JavaScript inside HTML
External JS	JavaScript stored in a separate .js file
Comments	Used to explain code

Unit 2: Variables and Data Types

Variables

Definition

A **Variable** is a named memory location used to store data.

Syntax

```
keyword variableName = value;
```

Example

```
let name = "Hari";  
  
document.write(name);
```

Output

Hari

var Keyword

Definition

var is the old way of declaring variables in JavaScript.

It can be redeclared and updated.

Example

```
var name = "Hari";  
  
document.write(name);
```

Output

Hari

let Keyword

Definition

let is used to declare variables that can be updated but cannot be redeclared in the same scope.

Example

```
let age = 20;

age = 21;

document.write(age);
```

Output

21

const Keyword

Definition

const is used to declare constant variables.
Its value cannot be changed after initialization.

Example

```
const college = "Sri Sairam Engineering College";

document.write(college);
```

Output

Sri Sairam Engineering College

Difference Between var, let and const

Feature	var	let	const
Can be Reassigned	✓	✓	✗
Can be Redeclared	✓	✗	✗
Block Scope	✗	✓	✓

Data Types

Definition

A **Data Type** specifies the type of value stored in a variable.

1. Number

Stores numeric values.

Example

```
let marks = 95;  
  
document.write(marks);
```

Output

95

2. String

Stores text values enclosed in quotes.

Example

```
let name = "Harikrishnan";  
  
document.write(name);
```

Output

Harikrishnan

3. Boolean

Stores either **true** or **false**.

Example

```
let result = true;  
  
document.write(result);
```

Output

true

4. Undefined

A variable that is declared but not assigned a value has the value **undefined**.

Example

```
let x;  
  
document.write(x);
```

Output

undefined

5. Null

Represents an empty or unknown value.

Example

```
let data = null;  
  
document.write(data);
```

Output

null

6. Object

Stores data as **key-value pairs**.

Example

```
let student = {  
  
  name: "Hari",  
  
  age: 20,  
  
  department: "IT"
```

```
};
```

```
document.write(student.name);
```

Output

Hari

7. Array

Stores multiple values in a single variable.

Example

```
let subjects = [
```

```
"HTML",
```

```
"CSS",
```

```
"JavaScript"
```

```
];
```

```
document.write(subjects[0]);
```

Output

HTML

Summary of Data Types

Data Type	Example
Number	100
String	"Hari"
Boolean	true
Undefined	let x;
Null	null
Object	{name:"Hari", age:20}
Array	["HTML", "CSS", "JS"]
